

Integrated Barometric Pressure Sensors on Legged Robots for Enhanced Tactile Exploration of Edges

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Abstract—This letter presents a new tactile sensor that utilizes an array of barometric pressure sensors encased in a deformable rubber sphere. Designed as an end effector foot of the legged quadruped robot Unitree Go1, the proposed sensor is able to withstand repeated impacts of at least 40 N at the feet. Tactile sensing in legged robotics extends their utility specifically in the context of edge detection and exploration. The presented tactile contact framework processes the pressure data to classify the type of contact (no contact, flat or edge) and the orientation of the edges relative to robot base. To assess the performance of the sensors and their ability for tactile edge exploration we extensively test with the legged robot in varied conditions including different terrains, changing payloads, exposure to dynamic disturbances, and sloped edges. The edge detection is compared against the original scalar force sensors. Experiments demonstrate a mean absolute error of 2° for predicted edge orientation at a detection range of 14 mm and robust operation in realistic operating conditions for medium-sized quadruped robots. With this contribution, we aim to enhance the capabilities and safety of legged robots in various applications.

Index Terms—Legged robots, soft robot applications, soft sensors and actuators.

I. INTRODUCTION

QUADRUPEDAL robots are becoming more versatile in traversing diverse terrains and environments [1], [2], [3]. They are already deployed for routine tasks such as inspection and automated data collection in environments including industrial plants, construction sites and factory floors. Nonetheless, these environments can be difficult to navigate due to various factors including staircases, cluttered spaces, and obstacles on the ground, which make it challenging to find suitable foot placements [4].

The safe operation of legged robots depends on their ability to perceive the terrain, which can be achieved using range sensors like Lidar [5], time-of-flight [6] or stereo RGB-D cameras [7].

Manuscript received 24 January 2024; accepted 11 May 2024. Date of publication 24 May 2024; date of current version 4 June 2024. This letter was recommended for publication by Associate Editor J. Qu and Editor C. Laschi upon evaluation of the reviewers' comments. This work was supported by Flemish Government under the "Onderzoeksprogramma Artificial Intelligence (AI) Vlaanderen" programme and Research Foundation – Flanders (FWO) under Grant 12Z4722 N (A.C.) and Grant S007723 N. (Corresponding author: Thijs Van Hauwermeiren.)

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This letter has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2024.3405378>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2024.3405378

These sensors enable the robot to create a map of the surrounding terrain, which can then be processed into a binary foothold score map to identify valid and invalid positions [8], [9]. However, range sensors can become occluded resulting in blind spots and uncertainty at the foothold [10], [11]. While perception based on range sensors is useful for planning, reactive control which can react to unsupportive contact is not easily realized.

A different approach to perceive the environment is to measure contact directly in the end effector feet, which can be achieved using tactile sensors [12]. Unlike range sensors, tactile sensors are in direct contact with the ground surface, even in situations where it might be obscured by clutter. Tactile sensors possess the ability to sense and localize forces, detect contours, and terrain features [13], [14]. In previous studies, tactile exploration on common legged robots weighing over 10 kg was not demonstrated, potentially due to limitations posed by the low admissible force levels of the utilized tactile technology.

In this work we introduce a reactive legged robot capable of exploration on the edge using barometric pressure sensors. Motivated by our prior work, where we noted the ability of barometric sensors to withstand high forces [15], we designed a deformable tactile foot sensor with high admissible forces typically occurring during contact on a medium sized legged robot. The key aspect of our approach is the use of deformation in the sole of the foot sensor to extend the sensing area and capture information about the terrain features. Moreover, we created a framework that can accurately estimate contact features such as foot contact and the direction of the edge from a single contact. The presented sensor solution consists of offline feature learning on 3D printed test objects. During deployment the quadruped robot can react to edges in the terrain, effectively enabling tactile exploration.

The sensor and models are validated during tactile exploration experiments under diverse conditions (changing payloads, dynamic disturbances, sloped edges) with the sensors fitted on the feet of the legged robot Unitree Go1 (Fig. 1).

In summary, our contributions are as follows:

- Design of a soft tactile foot for a legged robot based on barometric pressure sensors embedded in a rubber sphere.
- A tactile contact framework that can detect and classify types of contacts. The framework is able to accurately determine the orientation of edge contacts relative to the end effector frame.
- Experimental validation the proposed tactile sensor and models on the legged robot during exploration. The experiments are executed under various conditions (different

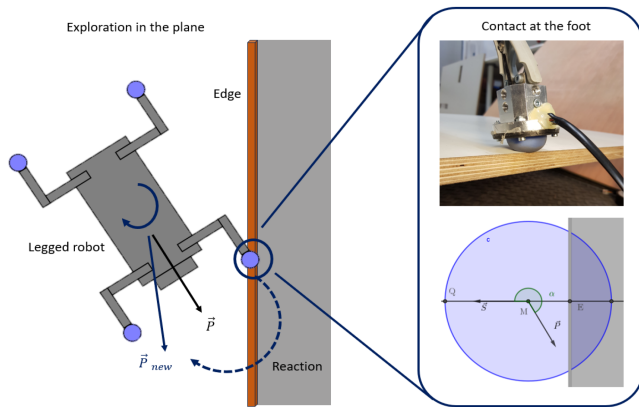


Fig. 1. Tactile exploration on legged robot. Barometric pressure sensors are integrated in the feet of the robot. Using a tactile contact framework to estimate the angle α between the robot orientation $\vec{P}$ and edge $\vec{S}$ the robot can adapt its pose when an edge is detected.

terrains, changing payloads, sloped edges, dynamical disturbances) together with a determination of its prediction accuracy and sensing range.

II. RELATED WORK

Tactile sensors, available in various types, serve multiple purposes, including edge detection [16], [17], force measurement [18], force localization [15], terrain classification [19], tactile servoing [20], [21], tactile exploration [13], [22], [23], in-hand manipulation [24] and slip detection [25]. Tactile sensing relies on physical principles such as optical, barometric, magnetic, and piezoresistive, each offering unique advantages and facing specific limitations [26], [27].

Optical tactile sensors that register the displacement of markers with a camera like TacTip [28], GelSight [29] and GelSlim [30] are often used in tactile grippers due to their high sensitivity and resolution however their typical applied force range is no more than 15 N. In [12] a vision based tactile sensor was used in single leg of a small crawling robot for exploration. For locomotion applications of larger sized legged robots, the primary design criteria is force and impact resistance.

Tactile sensors based on barometric MEMS sensors are typically embedded in a casted rubber form [15], [31], [32], [33], [34]. The rubber serves dual purposes: it protects the sensor and acts as a medium, channeling dispersed contact forces into measurable pressure. Previous designs demonstrated a low operational force range of 3N [34]. In our prior work [15] we noted the ability of higher force withstanding at least 25 N in gripping application provided the rubber is degassed during casting to ensure direct contact with the MEMS [35].

Slip prevention is also studied in the context of safe tactile operation of legged robots. Slipping occurs when insufficient friction is available for the ground reaction forces (GRF) causing the foothold to move despite being in contact. Force based slip detection relies on sensing the GRF [36]. Other methods for slip detection exist, for example IMU based [37]. [38] used force/torque sensors near the feet of a legged robot for terrain classification.

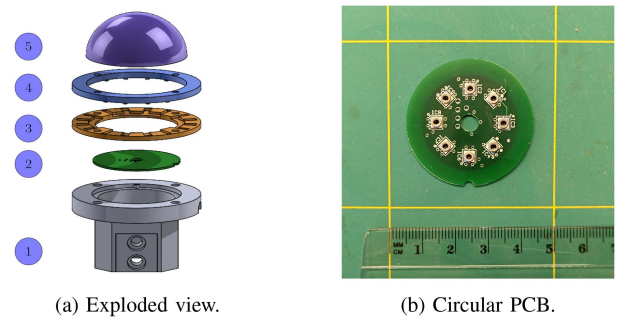


Fig. 2. (a) Components of the tactile end effector foot: 1) sensor holder, 2) PCB, 3) flow ring, 4) restraining ring and 5) rubber half sphere. (b) Detail of PCB showing the circular distribution of individual barometric pressure sensors.

In this work we design and deploy tactile end effectors feet on a legged robot based on an array of barometric pressure sensors encased by a rubber half sphere. This letter builds upon related research on barometric sensors integrated in the foot of legged robots [31], [32], which focused on sensing reaction forces. Our proposed sensor design has another distribution of MEMS elements for optimized sensitivity for the novel application of edge detection combined with high force capability.

III. TACTILE BAROMETRIC SENSOR DESIGN

The tactile sensor is comprised of eight barometric pressure sensors (MS5840-02BA from TE Connectivity) based on micro-electromechanical systems (MEMS) technology. We refer to an individual barometric pressure sensor as *sensor element*. The eight sensor elements are spatially arranged in a circular pattern (see Fig. 2(b)) using a custom designed Printed Circuit Board (PCB). Each element provides digital pressure readings. Serial communication is realized via **I2C with a multiplexer (TCA9548 A from Texas instrument)**, both sensors and multiplexer are integrated on the four layer custom PCB. The assembly has a **sampling frequency of 25 Hz**. A single four wire cable connects the sensor multiplexer to a micro-controller (PJRC Teensy 4.0) outside which processes the data stream. **Eight devices is the maximum number a single I2C multiplexer can connect**. Thus the eight barometric sensors is a compromise between form factor of PCB, latency of data processing and obtainable accuracy.

The mechanical design of the sensor is depicted in Fig. 2(a) and features an aluminum machined sensor holder, two 3D printed rings, a rubber casted sphere and embedded PCB. The rubber sphere is designed in a way that it closely resembles the original end effector foot in shape. The sensor holder houses the PCB and allows to fixate the sensor to a robotic end effector. The flow ring (orange) prevents the pcb from rotating and allows rubber to flow inside small canals. During casting the rubber flows to the bottom of the sensor holder such that the PCB is fully encapsulated into the silicon rubber. The restraining ring (blue) keeps the rubber sphere in place during deployment. A mix of **two component silicon rubber (Rision Shore index 15 A)** is poured into prepared cast, degassed and left to cure for 12 hours.

The force withstanding capability of the MEMS-based barometric tactile sensor depends on two factors: firstly, the peak pressures within the rubber, capped by the MEMS sensor's maximum pressure tolerance; and secondly, the rubber's maximum deformation capacity, which depends on its stiffness and the shape into which it's molded. The MS5840-02BA has an absolute maximum pressure rating of **10 bar**. The sensor was pressed on a glass plate and measuring the deformation with a camera underneath. A measured contact diameter of 22 mm was noted, equivalent to 379 mm², while a force of 100 N was applied measured with a ATI F/T sensor. At nominal force of 40 N, the measured contact diameter is 14 mm.

The sensor is designed to fit as a replacement foot for the Unitree Go1 quadruped robot (weight of 12 kg, 12 servo motor controlled joints, default height 0.34 m). The material cost of one sensor is approximately 140\$ including machined parts (80\$), PCB (10\$) and barometric sensors elements (50\$).

IV. TACTILE CONTACT FRAMEWORK

A. Contact Classifier Model

The sensor assembly signal $\mathbf{S}(t)$ sampled at time instant t consists of an array of eight pressure signals elements s_i .

$$\mathbf{S}(t) \in \mathbb{R}^8 = [s_1 \ s_2 \ \dots \ s_8]^T. \quad (1)$$

The objective is to classify contact situations in three categories: 1) no contact, 2) flat surface contact and 3) edge contact with the latter further characterized by: orientation relative to the sensor and its distance from the edge. The classification can be done by extraction of statistical features on the stream of data: mean $\mu(t)$ and standard deviation $\sigma(t)$ of $\mathbf{S}(t)$, calculated at each time instant t . We use the coefficient of variation (CV) representing the standard deviation normalized to the mean to classify the contact, with the time dependency dropped in notation for brevity:

$$CV = \frac{\sigma}{\mu}. \quad (2)$$

Practical experience gained during deployment revealed wide variance in magnitude of pressure signals, caused by impacts of making and breaking of contact. This led us to develop a hybrid approach, utilizing statistical features (which do not suffer from scaling issues) for robust classification of contact types discussed in this subsection and a neural network regressor for edge orientation discussed in Section IV-B. Both surface and edge contacts match the condition $\mu \geq \mu_{th}$. Edge contacts additionally satisfy the condition $CV \geq CV_{th}$. The sensor operation in time domain as the sensor makes contact around $t = 1.5$ s and breaks contact at $t = 4.5$ s is illustrated in Fig. 3 which shows the full sensor deformation in Fig. 3(a) and (b), measured pressure signals and CV plotted as function of time for each sensor element in case of a full surface contact (Fig. 3(c) and (e)) and an edge contact (Fig. 3(d) and (f)) on a 3D printed test object. The non-zero measured pressure before contact is measured residual pressure in the rubber. When the sensor is on an edge, a clear deviation between the signals coming from the sensor elements becomes apparent, while in the full contact

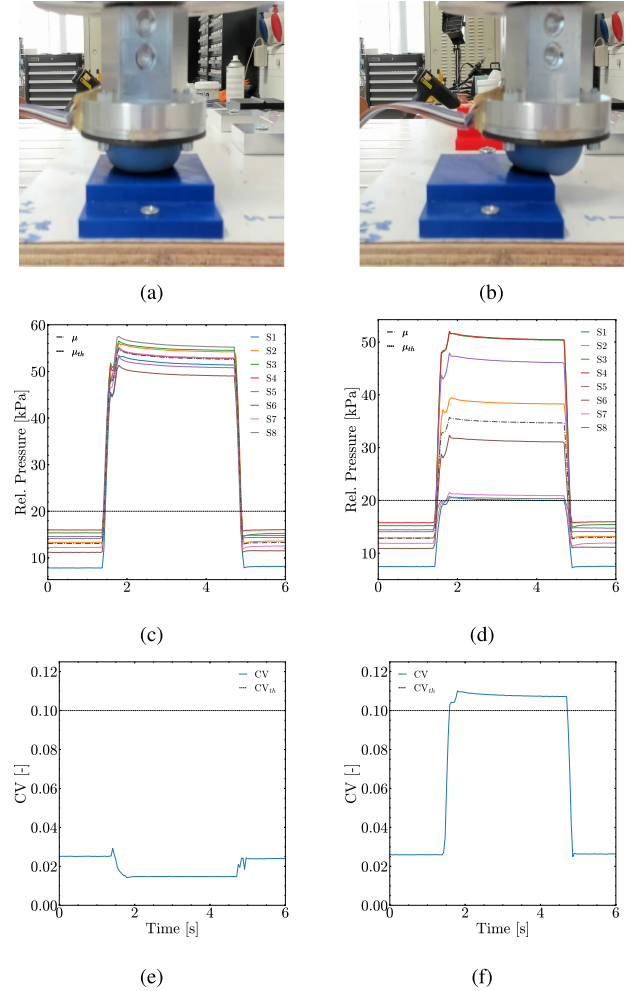


Fig. 3. Tactile sensor operating principle: Left column: surface contact. Right column: edge contact. Top row: sensor (mounted on UR5 robot) in contact. Middle row: barometric pressure signal $\mathbf{S}$ and mean μ . Bottom row: CV.

the signals are closer to each other. Also visible in Fig. 3(b) as opposed to Fig. 3(a) is the stretching of the rubber sole of the sensor around the edge, which causes the difference in measured pressures.

In Fig. 3 the sensor is in an upright position and has a zero pitch angle θ . The ability of the sensor to distinguish between surface and edge contacts depends on the pitch angle θ and the choice of the parameter CV_{th} . The sensitivity analysis is depicted in Fig. 4. The experiment reuses the same setup as the experiment in Fig. 3 and retains the 3s of contact data to form the box plot. The pitch angle is adjusted in the range $\theta = (-35, 35)^\circ$ while the contact location is the same. Based on a threshold value for edge classification $CV_{th} = 0.10$ the surface contact is correctly classified in the range $\theta = (-15, 15)^\circ$, with false positive edge detection outside that range (Fig. 4(a), (b)). Equally, the edge is correctly detected in the range $\theta > -5^\circ$ with false negative detection outside that range (Fig. 4(c), (d)). The value of CV_{th} is thus chosen as a compromise between precision and recall of edge contact classification. Given the circular symmetry of the sensor these results are also valid for

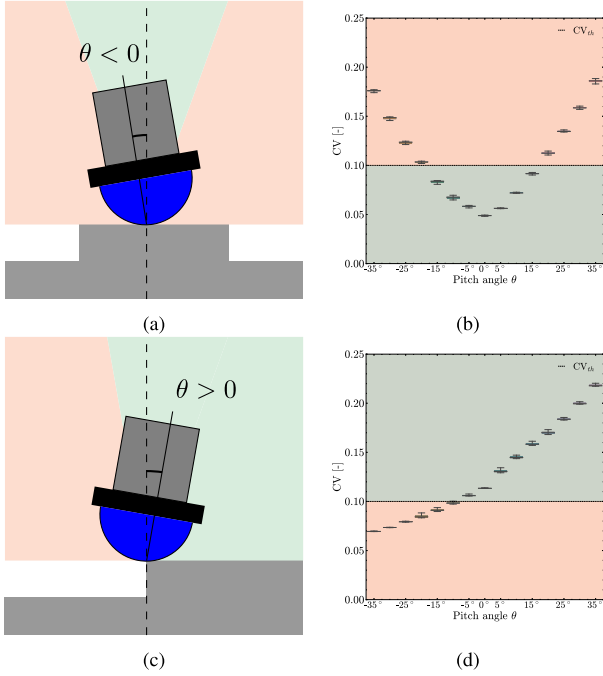


Fig. 4. Sensitivity analysis of the sensor operation on the pitch angle θ and threshold CV_{th} . Green area indicate correct classification, red area indicates incorrect classification. (a, b) Flat contact. (c, d) Edge contact.

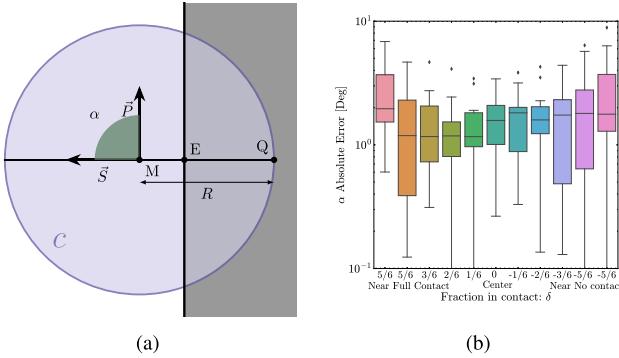


Fig. 5. Prediction error of edge orientation $\hat{\alpha}$ depending on the fraction of the extended rubber surface δ in contact with the edge of the terrain. (a) Model definitions based on a topside sketch of sensor in edge contact, the dark grey rectangle represents solid surface, the blue circle the contact circle of the sensor. (b) Results of sensitivity analysis.

B. Edge Regression Model

In this subsection we introduce the regression model $\mathcal{M}$ that characterizes the edges based on the sampled sensor readings $\mathbf{S}$.

Edge contacts are defined on the basis of the following definitions Fig. 5(a): Circle c represents the contact surface of the sensor with M the center. $\vec{P}$ is the heading, the orientation of the sensor frame relative to the robot frame and $\vec{S}$ the direction of stable support. The point E is defined as the perpendicular projection from M on the edge. Point Q is the intersection of the circle c with line ME , in the direction of $\vec{S}$. The angle α is the orientation of the edge relative to the heading. It is defined as the angle between $\vec{P}$ and $\vec{S}$ measured counterclockwise. Thus the sensor is able to differentiate between parallel edges with

opposite sides that give support. $R = 7\text{mm}$ is defined as the radius of the contact circle:

$$\delta = \frac{\|QE\| - R}{R} \in [-1, 1]. \quad (3)$$

The distance variable $\delta = 0$ if around 50% of the contact patch is in stable contact, $0 < \delta < 1$ if more than half of the contact surface is stable and vice versa $-1 < \delta < 0$ if less than half. The boundary conditions where the contact transitions from edge to surface and edge to no contact are represented respectively by $\delta = 1, -1$. The edge characterisation model $\mathcal{M}$ is a regression consisting of a fully connected neural network model which takes as input an eight dimensional pressure signal ($\mathbf{S}$) and gives as output α_x, α_y the cosine and sine of α , respectively. This intermediate step is necessary as α represents a circular variable. Assuming the heading $\vec{P}$ can be inferred from the robot IMU and forward kinematics using the joint state of the leg, the angle α can be used to locate the orientation of edges and stable surfaces. The neural network is fully connected and consists of an input layer, two hidden layers and output layer with dimensions (8, 12, 25, 3) nodes and activation functions (ReLU, ELU, sigmoid) respectively. The networks are trained for 200 epochs using Tensorflow/Keras using the SGD optimizer and a learning rate of 0.3. In addition, all input data is scaled in the range $[0, 1]$ before processing by the neural net.

The training data for $\mathcal{M}$ are generated by mounting the tactile sensor on the end-effector of a UR5 cobot (Universal Robots), together with an ATI Force/Torque (F/T) sensor. The UR5 provides a ground truth for position and orientation while the ATI F/T sensor provides the ground truth for forces. Data is then collected under controlled conditions in the following way: the end effector of the UR5 is moved from above the plane in a downward motion towards the ground through position control until a nominal force of 40 N is measured, which is the expected static load in deployment conditions. No force is applied on the end effector as it moves in the plane. These cycles, which we call pokes, are executed so they replicate the actual conditions of making and braking contact on a legged robot.

A working table in front of the UR5 provides the training part with known, regular shape. The location on the edge $\delta \in (-1, 1)$ is discretized in 8 different increments and orientation $\alpha \in (0, 2\pi)$ in 32 steps (8 steps on four sides). The combination results in 256 unique poking positions representing the different edge contacts. The full training database consists of 256 pokes for edge contacts and an 256 pokes of flat surface and no contact sampled near the edge. No data from irregular contacts such as corners is added to the training database.

The prediction accuracy of the predicted variables $\hat{\alpha}$, edge orientation, is quantified using a validation dataset. ($\delta \in (-1, 1)$ is discretized in 11 steps and orientation $\alpha \in (0, \pi)$ in 13 steps). This data does not correspond to data points in the training data, but is in the same distribution. Specifically, we investigate how the robustness of prediction depends on position on the edge: absolute error on the orientation $|\hat{\alpha} - \alpha|$. The resultant error plots are in Fig. 5(b). Overall, the Mean Absolute Error (MAE) for edge orientation α is the lowest between the middle of the edge and the full surface contact ($\delta = 2/6$, 75% quartile below

2° error). As more sensors are activated accuracy improves. However, as the sensor moves away from the middle of the edge ($\delta = 0$) towards the boundaries of edge contact ($\delta = \pm 1$), the variation and thus information present in the signal, drops. The median prediction error is below 2° for all positions. Outliers are present on all positions, with largest systematic errors furthest from the middle of the edge.

V. EXPERIMENTS ON QUADRUPEL ROBOT

The standard scalar force sensor of the unitree Go1 are replaced with our developed tactile sensors. The Go1 is position controlled during all experiments. Setpoints for joint positions are provided at 1 kHz. Additionally a custom walking controller and state machine is implemented using Bézier curves providing control of the base on the plane: walking forward, backward and turning. The legged robot is restricted to blind locomotion and tactile information only, it does not use camera feedback or other perception of the environment for walking.

The three-dimensional cartesian position (x, y, z) and orientation (quaternion) of the base of the Go1 is tracked using visual slam with a stereo RGB-Depth camera (ZED2 from Stereolabs). The camera is firmly attached to the robot base near the center of mass (CoM) and fuses the stereo RGB-Depth, IMU and magnetometer data stream to the desired pose output. This transform is provided by the position tracking API provided by Stereolabs.

The robot is moving towards a goal by an operator selecting a discrete action (e.g. walk forward, turn left, turn right). The walking controller is implemented as an alternating trot such that the diagonal pair of legs (FR, RL and FL, RR) alternate as swing and stance legs, with a stable standing in between. This allows for a fine-grained position control of the legged robot and plays an essential role in the presented experiments. Each experiment is recorded and a video clip is made available as supplementary material. During all experiments the thresholding parameters are $\mu_{th} = 20$ kPa (relative) and $CV_{th} = 0.1$.

A. Baseline Comparison

In this section we perform a comparison between the standard scalar force sensor (which provides a force measurement at the feet) originally equipped on the Go1, with our developed tactile sensors. The Go1 is commanded to walk towards the edge (Fig. 6(a)). The edge cannot be detected from the measurement data in Fig. 6(b)), resulting in a sudden drop of the robot without prior indication (frame 1a and 1b in Fig. 6(a)). The same locomotion is repeated with the tactile feet: once on the same wooden base (frame 2 Fig. 6(a)) and once on stone bricks which have a more rough edge and texture (frame 3 Fig. 6(a)). On both terrains the edge is detected (CV above threshold, Fig. 6(c) and (d)) with the robot standing stable on the edge.

B. Tactile Exploration

The legged robot moves from a start pose towards a goal pose in the plane with an carved out edge as obstacle in between. The robot is commanded to walk forward until an edge is

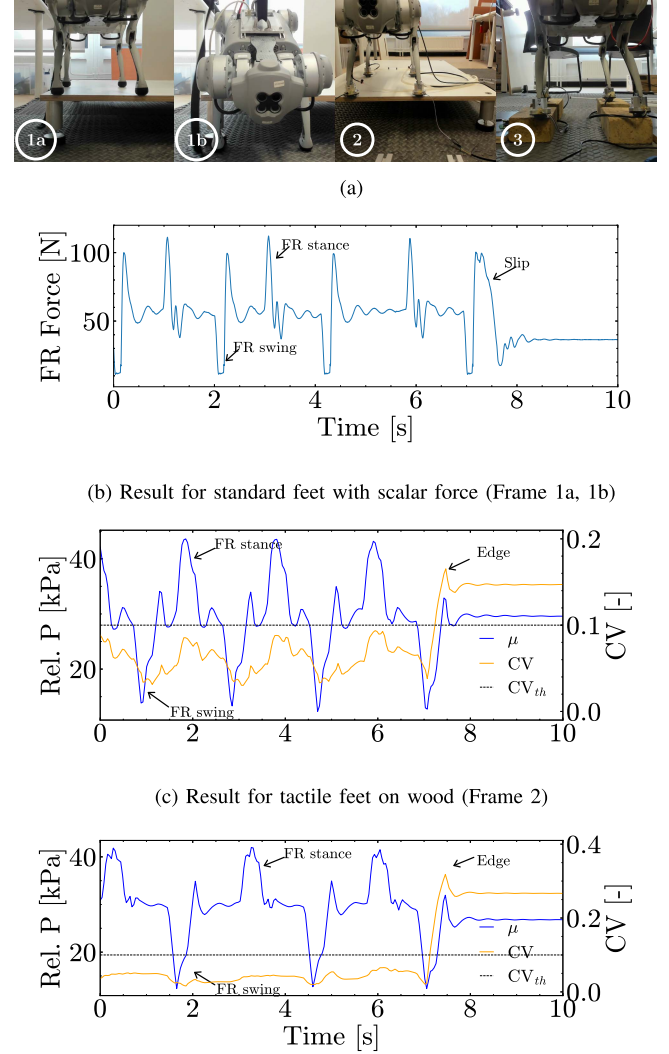


Fig. 6. Baseline experiment. (a) Standard feet frame 1a, 1b results in plot (b). Tactile feet on wood frame 2 results in plot (c). Tactile feet on stone brick frame 3 results in plot (d).

detected. When one of the front feet detects an edge, the robot is command to step back, turn and follow a deviated path. In the experiment depicted in Fig. 7 the edge of the surface is sensed multiple times by the front feet, resulting in a deviated direction of 90 degrees. Solely relying on the tactile information at the feet, the robot travels beyond the obstacles supported by a simple reactive path planning. The numerical results for this experiment are summarized in Table I including all sensed edges. The robot orientation (yaw) is obtained from the ZED camera IMU. α , the ground truth value for the edge angle is function of the yaw angle and $\vec{S}$, the vector of stable support (i.e. the orientation of the edge). $\hat{\alpha}$ is the prediction out of the neural network regression model $\mathcal{M}$ and $\Delta = |\alpha - \hat{\alpha}|$. It is noted that the maximum registered deviation in orientation error on the UR5 by a margin of 3°–5°, depending on position. This difference is explained due to the floating base of the legged robot, such that contacts for all legs are coupled and forces are distributed between contacts. This coupling effect

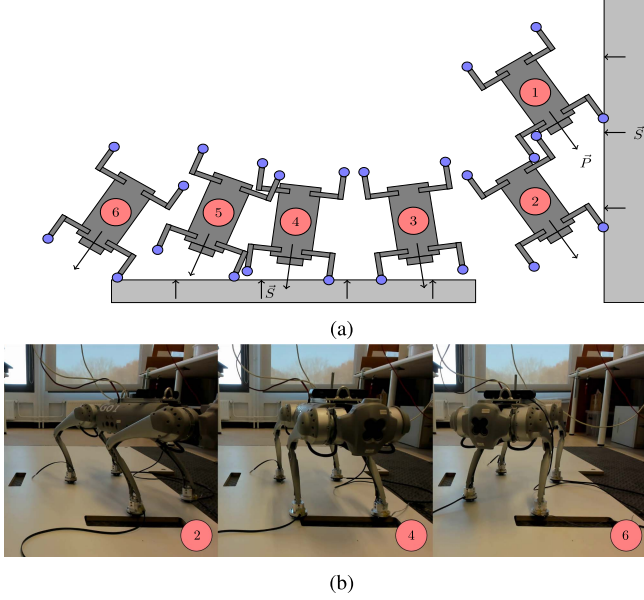


Fig. 7. Tactile exploration on Unitree Go1. (a) Schematic overview of detected edges (grey surfaces), not to scale for visualisation purposes. (b) Selected snapshots of corresponding frames.

TABLE I
NUMERICAL RESULTS FROM THE TACTILE EXPLORATION EXPERIMENT (FIG. 7)

Frame	$\vec{S}$	Leg	Yaw [rad]	α [rad]	$\hat{\alpha}$ [rad]	Δ [deg]
1	$\leftarrow$	FL	0.48	-2.05	-2.07	1.52
2	$\leftarrow$	FL	0.47	-2.04	-2.08	2.20
3	$\uparrow$	FR	0.24	2.90	3.02	7.20
4	$\uparrow$	FL	0.04	3.10	2.90	11.43
5	$\uparrow$	FL	-0.22	-2.92	-2.79	7.36
6	$\uparrow$	FL	-0.48	-2.67	-2.46	11.59

Each row corresponds with a detected edge (frame 2, 4, 6 represented by a picture in Fig. 7).

results in vibrations which make sensor data noisy and thus edge orientation regression on the legged robot more challenging, compared to the conditions during model training.

C. Payload Disturbance

One of our key differences with state-of-the-art sensors, is the ability of our sensors to withstand higher forces allowing for a more thorough tactile exploration of the environment. To validate their robust performance under varying loads, additional weights (copper brass disks each weighing approximately 0.75 kg) are firmly fixated near the center of mass (Fig. 8). The robot is commanded to walk forward until the edge is detected, then the robot steps backward, an additional weight is added, and the experiment is repeated. Up to five weights (3.75 kg or 30% robot mass in total) were added without sensor failure. The functioning of the sensor itself and the impact on the data processing is observed in Fig. 9. From these plots is observed that the relative pressure rises as weights are added. Also visible is that the CV raised during surface contacts as more payload is

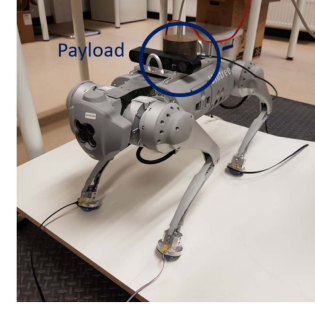


Fig. 8. Setup for payload disturbance experiment. Up to 30% additional robot mass is added.

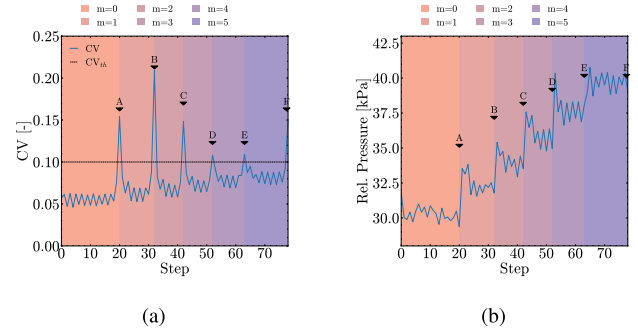


Fig. 9. Static payloads experiment: At the reference frames (A $\rightarrow$ F) the edge is detected and an additional payload is attached. Background color indicates the payload. (a) CV. (b) Relative pressure.

TABLE II
NUMERICAL RESULTS FROM PAYLOAD EXPERIMENT

Frame	m	Leg	Yaw [rad]	α [rad]	$\hat{\alpha}$ [rad]	Δ [deg]
A	0	FL	-0.17	-2.97	-3.00	1.76
B	1	FL	-0.17	-2.97	-3.05	4.52
C	2	FL	-0.18	-2.96	-3.00	2.45
D	3	FL	-0.19	-2.95	-2.91	2.23
E	4	FL	-0.24	-2.90	-2.72	10.34
F	5	FL	-0.29	-2.85	-2.93	4.28

Each row corresponds with a detected edge in Fig. 9.

added but the peak in CV remains detectable as the robot makes contact with an edge. From the results in Table II no increased errors in edge orientations is noted.

D. Inclined Slopes

The ability to detect not only straight edges but also inclined slopes can increase the applicability of the tactile sensor foot and also shows the robustness of the presented sensing approach. In the experiment the robot traverses an inclined slope, and this is several times repeated with different inclinations starting at co-inclination angle $\gamma = 15^\circ$ (near vertical) and ending at $\gamma = 75^\circ$ (near horizontal). The co-inclination angle γ is considered as the angle between the slope and the vertical direction, with $\gamma = 0$ being a straight edge. The test setup is depicted in Fig. 10

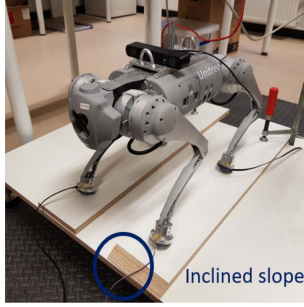


Fig. 10. Setup for inclined slope detection experiment. Five different floor pieces with different slopes are used.

TABLE III
NUMERICAL RESULTS FROM THE SLOPES EXPERIMENT

Frame	slope	Leg	Yaw [rad]	α [rad]	$\hat{\alpha}$ [rad]	Δ [deg]
A	15°	FL	-0.13	-3.01	-3.04	1.48
B	30°	FL	-0.04	-3.10	-3.13	2.00
C	45°	FL	-0.07	-3.07	-2.92	8.40
D	60°	FL	-0.02	-3.12	-2.94	10.55
E	75°	FL	0.05	-3.19	-3.10	5.06

Each row corresponds with a detected inclined edge in Fig. 11.

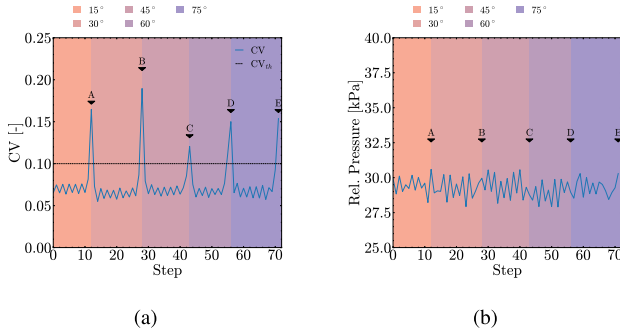


Fig. 11. Inclined slopes experiment: At the reference frames (A → E) the edge is detected on a different slope. Background color indicates the inclination of the slope. (a) CV. (b) Relative pressure.

and results are summarized in Table III. The main conclusion from this experiment is that different inclined slope edges can still be accurately detected on the basis of the processed data (CV) (see Fig. 11), up until at least $\gamma = 75^\circ$. The data shows no significant different pattern for edges with inclined slopes. This can be explained as in all investigated cases the sensor remains in contact with the horizontal section of the surface near the edge, regardless of inclination angle.

E. Dynamic Disturbance

The robot was pushed down several times by a cobot applying a controllable thrust during a 10 s time window showing its behaviour during a dynamic disturbance. The experiment was repeated for two scenarios: standing on the surface (Fig. 12(a), (c), (e)) and the edge (Fig. 12(b), (d), (f)). The plots shown are pressure signals (in kPa) and measured thrust (in N) (Fig. 12(c),

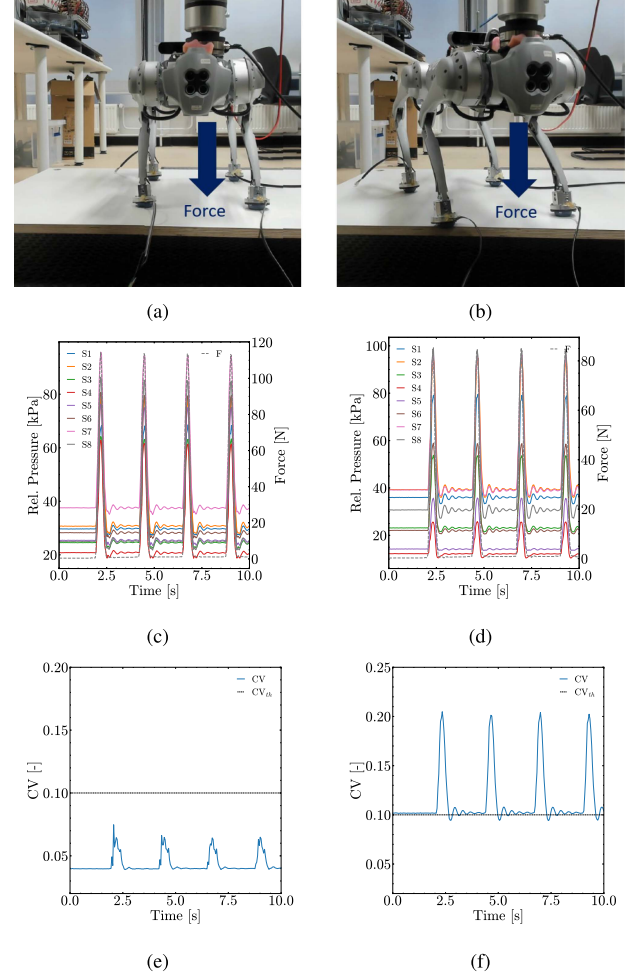


Fig. 12. Dynamic disturbance experiment: Legged robot pushed down several times by a cobot. Left column: robot standing on flat terrain. Right column: standing on near the edge.

(d)) and CV plots (Fig. 12(e), (f)). Pushes apply a thrust, measured by an ATI F/T sensor, of up to 100 N which cause a transient, the pressure values vary according to the mean, which is related to the applied thrust. During the transient, the classification principle remains mostly valid as is clear in the CV plots. After the transient pressure signals return to their original values. Hence, our sensors remain robust and accurate even under dynamic disturbances.

VI. CONCLUSION AND FUTURE WORK

In this study, we have developed and implemented a novel tactile foot sensor for the Unitree Go1, a medium-sized legged robot. This sensor enabled reliable tactile exploration of the legged robot across a wide variety of conditions. We introduced a tactile contact framework designed to determine the nature of the contact (whether surface, edge, or no contact) and the orientation of edges. Through comprehensive locomotion tests under varied conditions, including changes in load, on different terrains, dynamic disturbances, and interactions with sloped edges, the sensor displayed a worst-case edge orientation error

of 11°, with typical errors ranging between 2° and 7° with a detection range of 14 mm.

Notably, the sensor's robustness was exemplified during experiments simulating dynamic disturbances, where it remained fully operational under the robot's full weight, enduring additional forces up to 100 N. This resilience to impact forces is crucial for real-world applications.

In future work we will explore alternative sensor arrangements to improve the accuracy of contact state estimation further. Integrating tactile data with information from range sensors presents another promising avenue for enriching robot-environment interactions.

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